



**Generation Interconnection
System Impact Study Report
for
Queue Project AF2-088
SHAWVILLE-CLEARFIELD 34.5 KV II
3.9 MW Capacity / 6.5 MW Energy**

February 2021

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1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Mid-Atlantic Interstate Transmission, LLC (MAIT, PENELEC zone).

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

3 General

The Interconnection Customer has proposed an uprate to a planned/existing Solar generating facility located in Clearfield, Pennsylvania. This project is an increase to the Interconnection Customer's AF2-039 project, which will share the same point of interconnection. The AF2-088 queue position is a 6.5 MW uprate (3.9 MW Capacity uprate) to the previous project. The total installed facilities will have a capability of 20.0 MW with 12.0 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this uprate project is May 31, 2021. This study does not imply a TO commitment to this in-service date.

Queue Number	AF2-088
Project Name	SHAWVILLE-CLEARFIELD 34.5 KV II
State	Pennsylvania
County	Clearfield
Transmission Owner	PENELEC
MFO	20
MWE	6.5
MWC	3.9
Fuel	Solar
Basecase Study Year	2023

Any new service customers who can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

4 Point of Interconnection

4.1 Primary Point of Interconnection

AF2-088 will interconnect with the PENELEC distribution system at the same POI as AF2-039 which is a 34.5kV interconnection via a tap on the #2 Clearfield circuit at the Penelec-owned Shawville substation. The IC's proposed generating unit site is approximately 1.0 mile northwest of Woodland, PA., near 211 Independent Road. The IC will not incur any additional connection facility upgrade costs for AF2-088 as long as AF2-039 is constructed before or concurrently with AF2-088.

Attachment 1 shows a one-line diagram of the proposed primary direct connection facilities for the AF2-088 and AF2-039 generation projects to connect to the Penelec distribution system. IC will be responsible for constructing all of the facilities on its side of the POI, including the attachment facilities which connect the generator to the Penelec distribution system's direct connection facilities.

5 Cost Summary

The AF2-088 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$0
Total System Network Upgrade Costs (Summer Peak)	\$0
Total System Network Upgrade Costs(Light Load)	\$0
Total System Network Upgrade Costs (TO Identified)	\$0
Total Costs	\$0

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not

closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

Note 2: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

6 Transmission Owner Scope of Work

The AF2-088 will interconnect with the Penelec distribution system at the same POI as AF2-039 via a tap on the 34.5 kV #2 Clearfield circuit at pole # SBC-2-8021. The IC will be responsible for acquiring all easements, properties, and permits that may be required to construct the new interconnection station and the associated facilities.

Attachment 1 shows a one-line diagram of the proposed primary direct connection facilities for the AF2-088 generation project to connect to the Penelec distribution system. The IC will be responsible for constructing all of the facilities on its side of the POI, including the attachment facilities which connect the generator to the FE distribution system's direct connection facilities.

The IC will not incur any additional connection facility upgrade costs for AF2-088 as long as AF2-039 is constructed before or concurrently with AF2-088. If the AF2-039 project withdraws from the New Services Queue, then the AF2-088 project will be responsible for the interconnection facilities identified in the AF2-039 Facilities Study Agreement.

7 Schedule

This project is sharing the same attachment facilities with the AF1-039 project and will share the same schedule with that project if constructed at the same time.

If the customer is ultimately responsible for network upgrades, then the schedule for those upgrades will be refined in future study phases. The customer would need to wait for those upgrades to be completed prior to commercial operation unless determined deliverable by an interim deliverability study. The elapsed time to complete any network upgrades is provided in the System Reinforcements table of this report.

8 Transmission Owner Analysis

Penelec performed an analysis of its distribution system. The AF2-088 project did not contribute to any overloads on the distribution system for either the primary POI or the secondary POI.

9 Interconnection Customer Requirements

9.1 System Protection

The same analysis that was conducted to assess the impact of the Shawville-Clearfield 34.5 kV (AF2-039) Project on the system protection requirements in the area applies to the AF2-088 project. The results of this review show that the following relay additions will be required:

Proposed single line diagrams show Oriden, LLC (Developer) constructing a generation facility they call **“Abe’s Run”** tapping Penelec’s Shawville - 34.5kV #2 Clearfield circuit at pole SBC-2-8021.

The 34.5kV interconnection proposal will require Developer to meet applicable "Technical Requirements" as outlined in First Energy's document titled "Technical Requirements for the Interconnection of Customer-Owned Generation to the FirstEnergy Distribution System". Anti-islanding system shall meet IEEE 1547 and UL 1741 therefore no Direct Transfer Trip (DTT) will be required.

9.2 General Concerns

It is to be understood, for abnormal operation of the Penelec system, which could cause Developer’s generation facility to be electrically isolated from the Penelec system synchronous source via the tripping of an interconnecting primary voltage line or device, Developer will, via Penelec’s direction, be required to disconnect the generation from Penelec’s system and remain disconnected (**units are required to be OFF LINE**), until the Penelec system normal circuitry is restored. These abnormal conditions will be reviewed by Penelec system operators as to the need for the generation facility to be disconnected.

9.3 Requirements for Owner's/Developer's generation IPP Facility

The proposed interconnection Owner's/Developer's facilities must be designed in accordance with the document titled *FirstEnergy Distribution Engineering Practices Interconnection of Customer-Owned Generation to the FirstEnergy Distribution System* dated 11/17/14 located at the following link:

<http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>

The document is referred to as engineering practice EP(# 02-280) with section 4 part C specifically referencing the "interconnection technical requirements". Certain protection requirements are shown.

Additionally, Owner/Developer is responsible to provide adequate protection (for their equipment) under any distribution system operating condition' - which includes 'Separation from supply' (i.e. tripping of F.E. circuit breakers) and 'Re-synchronizing the generation after electric restoration of the supply' (i.e. reclosing of F.E. circuit breakers).

Owner's/Developer's protection must be designed to coordinate with the reclosing practices of FirstEnergy line protective devices. The generator must cease to energize the FirstEnergy circuit to which it is connected prior to reclosing of any (FE) automatic reclosing devices.

Owners/Developer's electrical protection and control schematics shall be provided to FE for consideration. FE may request modifications, if required, to meet the technical requirements.

9.4 Compliance Issues

Oriden, LLC will be responsible for meeting a power factor between 0.95 lagging (producing MVARs) to 0.95 leading (absorbing MVARs) and assure that voltage deviation will be less than 1.0 volt as measured at the POI under all Solar Gen operating conditions due to the inherent dynamic reactive power capability of this solar facility.

Generators with no inherent VAR (reactive power) control capability, or those that have a restricted VAR capability less than the defined requirements, must provide dynamic supplementary reactive support located at the generation facility with electrical characteristics equivalent to that provided by a similar sized synchronous generator. A Dynamic Reactive Compensation (either Static VAR Compensator (SVC) or STATCOM) or other method be applied in order to maintain the required specifications at the POI. Oriden, LLC is responsible for the installation of equipment on its side of the POI in order to adhere to the criteria stated above by FirstEnergy.

10 Revenue Metering and SCADA Requirements

10.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

10.2 Meteorological Data Reporting Requirements

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Back Panel temperature (Fahrenheit) - (Required for plants with Maximum Facility Output of 3 MW or higher)
- Irradiance (Watts/meter²) - (Required for plants with Maximum Facility Output of 3 MW or higher)
- Ambient air temperature (Fahrenheit) - (Accepted, not required)
- Wind speed (meters/second) - (Accepted, not required)
- Wind direction (decimal degrees from true north) - (Accepted, not required)

10.3 Interconnected Transmission Owner Requirements

The IC will be required to comply with all Interconnected Transmission Owner's revenue metering requirements for generation interconnection customers located at the following link:

<http://www.pjm.com/planning/design-engineering/to-tech-standards/>

11 Summer Peak Analysis

The Queue Project AF2-088 was evaluated as a 6.5 MW (Capacity 3.9 MW) injection as an uprate to AF2-039 at the Shawville 34.5 kV substation in the PENELEC area. Project AF2-088 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF2-088 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

11.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC/D C	MW IMPACT
99138131	200940	26SCOTCHHLLW	230.0	PENELEC	200675	26E.TWANDA	230.0	PENELEC	1	AP-P1-3-WP-230-326T	single	570.0	99.99	100.1	AC	0.64

11.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

11.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

11.4 Steady-State Voltage Requirements

To be determined in the Facilities Study phase.

11.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

11.6 System Reinforcements

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-088	Upgrade Number
99138131	1	26SCOTCHHLLW 230.0 kV - 26E.TWANDA 230.0 kV Ckt 1	PENELEC ProjectId : n7127.4 Description : Replace line trap at East Towanda. Type : FAC Total Cost : \$225,000 Time Estimate : 12 Months Ratings : 515.0/615.0/646.0 Queue Project AF2-088 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF2-088 could receive cost allocation. Note 2: Although Queue Project AF2-088 may not have cost responsibility for this upgrade, Queue Project AF2-088 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-088 comes into service prior to completion of the upgrade, Queue Project AF2-088 will need an interim study.	\$225,000	\$0	n7127.4
			TOTAL COST	\$225,000	\$0	

Note : For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

11.7 Flow Gate Details

The following indices contain additional information about each facility presented in the body of the report. For each index, a description of the flowgate and its contingency was included for convenience. The intent of the indices is to provide more details on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the indices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the indices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators. It should be noted the project/generator MW contributions presented in the body of the report are Full MW Impact contributions which are also noted in the indices column named "Full MW Impact", whereas the loading percentages reported in the body of the report, take into consideration the PJM Generator Deliverability Test rules such as commercial probability of each project as well as the ramping impact of "Adder" contributions. The MW Impact found and used in the analysis is shown in the indices column named "Gendeliv MW Impact".

11.7.1 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
99138131	200940	26SCOTCHLLW	PENELEC	200675	26E.TWANDA	PENELEC	1	AP-P1-3-WP-230-326T	single	570.0	99.99	100.1	AC	0.64

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200642	26SENECA#1	2.9113	80/20	2.9113
200643	26SENECA#2	2.7496	80/20	2.7496
200644	26SENECA#3	0.2674	80/20	0.2674
200649	26PENNTech	0.9584	80/20	0.9584
200665	26SHAWVL 3	5.9915	80/20	5.9915
200666	26SHAWVL 4	5.9242	80/20	5.9242
200715	26SHAWVL 1	3.8263	80/20	3.8263
200722	26SHAWVL 2	3.9052	80/20	3.9052
200898	26AA1-106	9.3427	80/20	9.3427
200905	26Q36	0.1688	80/20	0.1688
200913	26SHAW-D	0.1654	80/20	0.1654
201201	26WRREN CT	0.5862	80/20	0.5862
201477	26Y2-055	1.7768	80/20	1.7768
203261	26BLOSSCT	0.5405	80/20	0.5405
203351	26GROZ1-110	1.5699	80/20	1.5699
203352	26CANZ2-011	1.5699	80/20	1.5699
919201	AA1-144 OP	134.3185	80/20	134.3185
919491	AA2-000	183.6489	80/20	183.6489
930511	AB1-092	6.7430	80/20	6.7430
934801	AD1-108	0.1183	80/20	0.1183
934811	AD1-109	0.0868	80/20	0.0868
936421	AD2-055	13.8827	80/20	13.8827
936991	AD2-133 C	1.8077	80/20	1.8077
939171	AE1-147 C	2.7710	80/20	2.7710
940201	AE2-001 C	2.7653	80/20	2.7653
940681	AE2-055 C (Suspended)	2.6756	80/20	2.6756
941191	AE2-113 C	5.3197	80/20	5.3197
941261	AE2-120 C	2.7632	80/20	2.7632
941271	AE2-121 C	1.4756	80/20	1.4756
941321	AE2-126 C (Suspended)	1.7780	80/20	1.7780
941331	AE2-129 C	1.5487	80/20	1.5487
941351	AE2-131 C (Suspended)	1.5487	80/20	1.5487
941421	AE2-139 C	45.3848	80/20	45.3848
942351	AE2-248 C	2.1759	80/20	2.1759
942491	AE2-262 C	11.7510	80/20	11.7510
942501	AE2-263 C	11.0459	80/20	11.0459
942961	AE2-316 C	4.4988	80/20	4.4988
943751	AF1-043	41.6482	80/20	41.6482
944001	AF1-068 C O1 (Withdrawn : 12/15/2020)	2.8841	80/20	2.8841

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
944181	AF1-086 C O1	1.5308	80/20	1.5308
944311	AF1-099 C	17.8380	80/20	17.8380
944321	AF1-100 C	32.1825	80/20	32.1825
944381	AF1-103 O1	1.2240	80/20	1.2240
944471	AF1-112 C	2.9084	80/20	2.9084
944671	AF1-132 C O1 (Withdrawn : 12/15/2020)	2.6970	80/20	2.6970
944691	AF1-134 C O1	1.3556	80/20	1.3556
944771	AF1-142 C	28.5408	80/20	28.5408
944881	AF1-153 C O1	1.8592	80/20	1.8592
944901	AF1-155 C	1.8450	80/20	1.8450
945121	AF1-177	0.3060	80/20	0.3060
945161	AF1-181	0.1658	80/20	0.1658
945171	AF1-182	0.8247	80/20	0.8247
945181	AF1-183	0.1866	80/20	0.1866
945331	AF1-198	0.4233	80/20	0.4233
945491	AF1-214 C (Withdrawn : 12/03/2020)	2.7634	80/20	2.7634
946381	AF1-302 C	1.9645	80/20	1.9645
946421	AF1-306 C	8.1133	80/20	8.1133
957451	AF2-039 C	1.3290	80/20	1.3290
957941	AF2-088 C	0.6399	80/20	0.6399
958271	AF2-121 C	1.5487	80/20	1.5487
960041	AF2-295 C	1.8450	80/20	1.8450
960051	AF2-296 C	1.3556	80/20	1.3556
961461	AF2-437	-0.1124	Adder	-0.13
WEC	WEC	0.1676	Confirmed LTF	0.1676
LGEE	LGEE	0.2938	Confirmed LTF	0.2938
CPL	CPL	0.2097	Confirmed LTF	0.2097
CBM-W2	CBM-W2	3.9967	Confirmed LTF	3.9967
NY	NY	1.6784	Confirmed LTF	1.6784
TVA	TVA	0.6412	Confirmed LTF	0.6412
CBM-S2	CBM-S2	2.0750	Confirmed LTF	2.0750
CBM-S1	CBM-S1	3.9959	Confirmed LTF	3.9959
MEC	MEC	0.8183	Confirmed LTF	0.8183
CBM-W1	CBM-W1	6.9180	Confirmed LTF	6.9180

11.8 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

Queue Number	Project Name	Status
AA1-106	Grover II 34.5kV	Withdrawn
AA1-144	East Towanda-Grover 230kV	Engineering and Procurement
AA2-000	N/A	N/A
AB1-092	Moshannon-East Towanda 230kV	Active
AD1-108	Grover 34 kV	In Service
AD1-109	Canton 34.5 kV	In Service
AD2-055	Moshannon-East Towanda 230 kV	Active
AD2-133	Eagle Valley 115kV	Active
AE1-147	Bellefonte 46 kV	Engineering and Procurement
AE2-001	Nittany-Zion 46 kV	Active
AE2-055	Shingletown-Boalsburg 46 kV	Suspended
AE2-113	Farmers Valley-Ridgeway 115 kV	Active
AE2-120	Graymont-Zion 46 kV	Active
AE2-121	Milesburg-Tanney Junction 46 kV	Active
AE2-126	Dubois-Curwensville 34.5 kV	Suspended
AE2-129	Philipsburg-Clarence 34.5 kV	Engineering and Procurement
AE2-131	Philipsburg-Karthauss 34.5	Suspended
AE2-139	East Towanda-Grover 230 kV	Active
AE2-248	Fillmore-Thompson Farm 46 kV	Active
AE2-262	Moshannon-Milesburg 230 kV	Active
AE2-263	Moshannon-Milesburg 230 kV	Active
AE2-316	Brookville-Squab Hollow 138 kV	Active
AF1-043	Moshannon-East Towanda 230 kV	Active
AF1-068	Boalsburg-Centre Hall 46 kV	Withdrawn
AF1-086	Madera-Westover South 115 kV	Active
AF1-099	Moshannon-Milesburg 230 kV	Active
AF1-100	Shawville-Moshannon 230 kV	Active
AF1-103	Warren 34.5 kV	Active
AF1-112	Centre Hall-Boalsburg 46 kV	Active
AF1-132	Shingletown-Boalsburg 46 kV	Withdrawn
AF1-134	Philipsburg-Madera 34.5 kV	Active
AF1-142	Moshannon-Milesburg 230 kV	Active
AF1-153	Motion-Ridgeway 46 kV	Active
AF1-155	Paper City-Wilcox 46 kV	Engineering and Procurement
AF1-177	Warren 115 kV	Partially in Service - Under Construction
AF1-181	Shawville 3 230 kV	Partially in Service - Under Construction
AF1-182	Shawville 4 230 kV	Partially in Service - Under Construction
AF1-183	Shawville 1 230 kV	Partially in Service - Under Construction
AF1-198	Blossburg #1 CT 34.5 kV	Partially in Service - Under Construction

Queue Number	Project Name	Status
AF1-214	Nittany-Zion 46 kV	Withdrawn
AF1-302	Brookville-Squab Hollow 138 kV	Active
AF1-306	Squab Hollow 230 kV	Active
AF2-039	Shawville-Clearfield 34.5 kV	Active
AF2-088	Shawville-Clearfield 34.5 kV II	Active
AF2-121	Philipsburg-Shawville 34.5 kV	Active
AF2-295	Wilcox-Paper City 46 kV	Active
AF2-296	Madera 34.5 kV	Active
AF2-437	Oxbow 34.5 kV III	In Service
Y2-055	Elm Street 34.5kV	Withdrawn
Z1-110	Grover 34kV	In Service
Z2-011	Canton 34.5kV	In Service

11.9 Contingency Descriptions

Contingency Name	Contingency Definition
AP-P1-3-WP-230-326T	CONTINGENCY 'AP-P1-3-WP-230-326T' /* SHINGLETOWN #82 230/46KV XFMR DISCONNECT BRANCH FROM BUS 235248 TO BUS 236711 CKT 82 /* 01SHINGL 230 01SHINGLTN 46 DISCONNECT BRANCH FROM BUS 235248 TO BUS 200513 CKT 1 /* 01SHINGL 230 26LEWISTWN 230 DISCONNECT BRANCH FROM BUS 235248 TO BUS 235970 CKT 1 /* 01SHINGL 230 01DALE 230 DISCONNECT BUS 237058 /*SHINGLETOWN 230 KV CAPACITOR END

12 Light Load Analysis

Light load analysis not required for solar projects.

13 Short Circuit Analysis

The following Breakers are overdutied:

None

13.1 System Reinforcements - Short Circuit

None

14 Stability and Reactive Power

PJM does not require a stability analysis for this project.

15 Affected Systems

15.1 NYISO

None

16 Attachment 1: One Line Diagram

